

EBN111 Formulas

Chapter 1

Factor	Prefix	Symbol
10^9	giga	G
10^6	mega	M
10^3	kilo	k
10^{-2}	centi	c
10^{-3}	milli	m
10^{-6}	micro	μ
10^{-9}	nano	n
10^{-12}	pico	p

Quantity	Unit	Symbol
electric charge	coulomb	C
electric potential	volt	V
resistance	ohm	Ω
conductance	siemens	S
inductance	henry	H
capacitance	farad	F
frequency	hertz	Hz
force	newton	N
energy, work	joule	J
power	watt	W
magnetic flux	weber	Wb
magnetic flux density	tesla	T

$$e = -1.602 \times 10^{-19} \text{ C}$$

$$Q = \int_{t_0}^{t_1} i \, dt$$

$$i = dq/dt$$

$$1A = 1C/s$$

$$V_{ab} = dw/dq$$

$$V_{ab} = V_a - V_b$$

$$p = \frac{dw}{dt} = \frac{dw}{dq} \cdot \frac{dq}{dt} = vi$$

$$1 \text{ Wh} = 3600 \text{ J} = 3.6 \text{ kJ}$$

Chapter 2

$$R = \rho \frac{l}{\text{Area}} = \frac{V}{I}$$

$$V = IR$$

$$G = \frac{1}{R} \quad \text{or} \quad i = Gv$$

$$b = l + n - 1$$

$$\text{KCL:}$$

$$\text{KVL:}$$

b branches, **n** nodes, and **l** independent loops

$$\sum_{n=1}^N i_n = 0$$

$$\sum_{m=1}^M V_m = 0$$

Voltage Division:

Current Division (only 2):

Current Division (any):

$$V_n = \frac{R_n}{R_1 + R_2 + \dots + R_N} V$$

$$I_n = \frac{R_{n_other}}{R_1 + R_2} I$$

$$I_n = \frac{G_n}{G_1 + G_2 + \dots + G_N} I$$

$$R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$$

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \dots + \frac{1}{R_N}$$

Delta $\rightarrow$ Wye:

$$R_1 = \frac{R_b R_c}{(R_a + R_b + R_c)}$$

Wye $\rightarrow$ Delta:

$$R_a = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_3}$$

$R_a = R_b = R_c = R_\Delta$
and

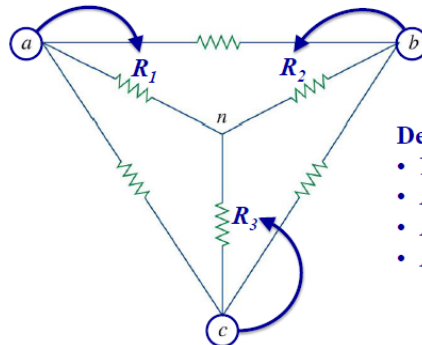
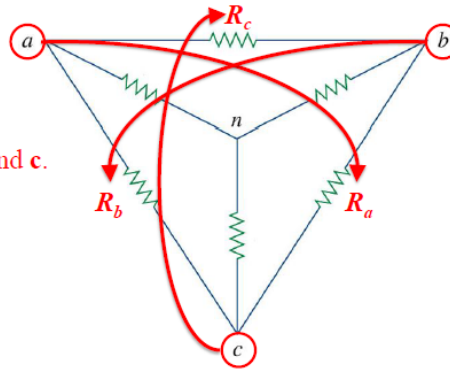
$$R_1 = R_2 = R_3 = R_Y$$

then

$$R_Y = \frac{R_\Delta}{3} \text{ or } R_\Delta = 3R_Y$$

Delta Δ : R_a, R_b, R_c .

- Name the nodes **a**, **b**, and **c**.
- R_a is opposite node **a**
- R_b is opposite node **b**
- R_c is opposite node **c**



Delta Y: R_1, R_2, R_3 .

- Name the nodes **a**, **b**, and **c**.
- R_1 is connected to node **a**
- R_2 is connected to node **b**
- R_3 is connected to node **c**

Chapter 3

Nodal analysis:

$$i = \frac{V_{High} - V_{Low}}{R_{Branch}}$$

$$i_{Branch} = \frac{V_{n_from} - V_{n_to}}{R_{Branch}}$$

Mesh analysis:

$$R(i_{loop} - i_{opposing_loop})$$

Transistors:

$$V_{CE} = V_{BE} + V_{CB}$$

$$I_E = I_C + I_B$$

$$V_{BE} = 0.7V$$

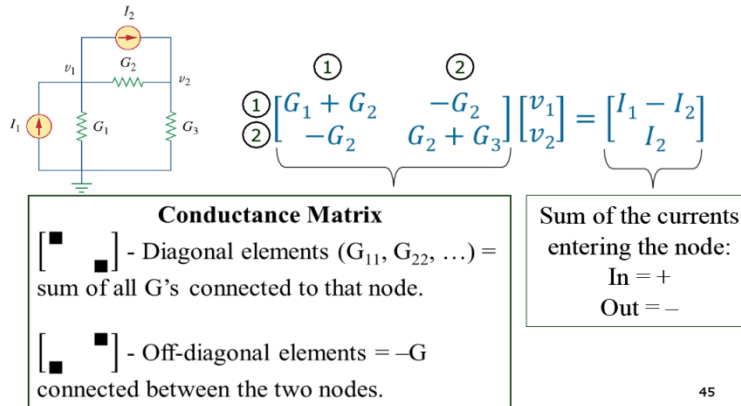
$$I_C = \beta I_B$$

$$\beta = \frac{\alpha}{1 - \alpha}$$

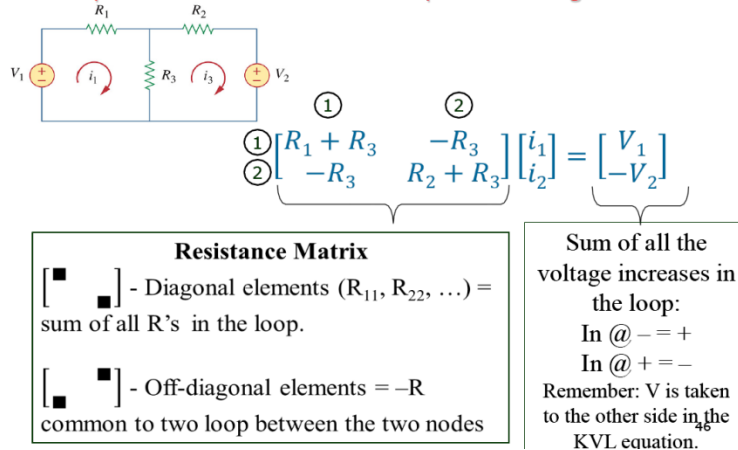
$$I_E = (1 + \beta)I_B$$

Nodal and Mesh Analysis by inspection:

Special case #1: All sources = independent current sources

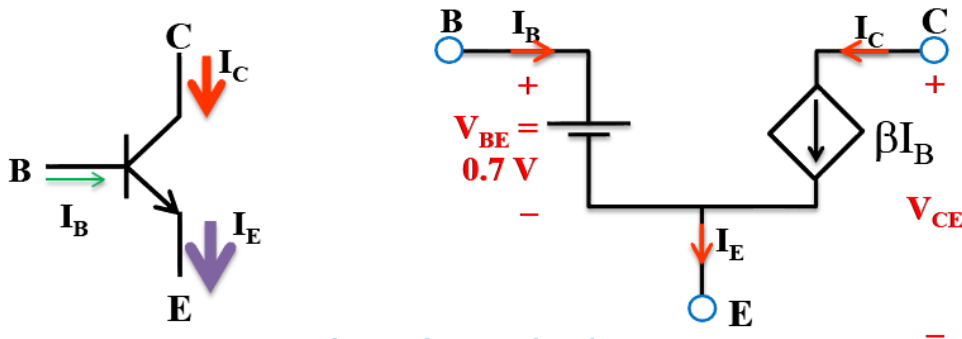


Special case #2: All sources = independent voltage sources



Transistors:

Equivalent network model



A transistor is equivalent to a current controlled current source.

Chapter 4

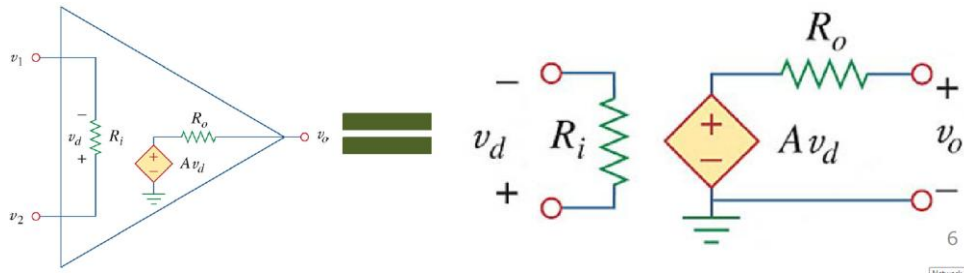
$$P = i^2 R_L = \left(\frac{V_{TH}}{R_{TH} + R_L} \right)^2 R_L$$

For max power:

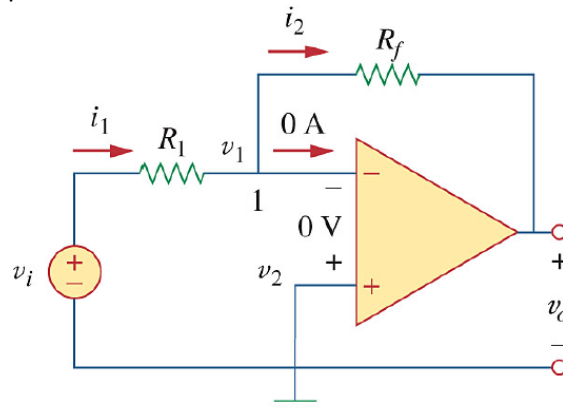
$$R_L = R_{TH}$$

$$P_{max} = \frac{V_{TH}^2}{4R_{TH}} = \frac{V_{RL}^2}{R_{TH}}$$

Chapter 5



Inverting Amplifier:



$$v_+ = v_- = 0V$$

KCL at v_-

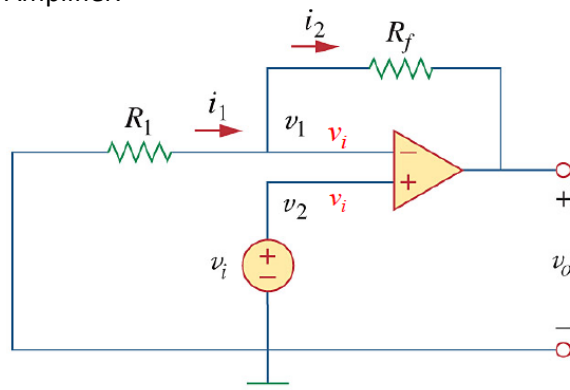
$$-i_1 + i_2 = 0$$

$$\frac{0 - V_i}{R_1} + \frac{0 - V_o}{R_f} = 0$$

$$\frac{V_o}{V_i} = -\frac{R_f}{R_1}$$

$$V_o = -\frac{R_f}{R_1} V_i$$

Non-inverting Amplifier:



$$v_+ = v_- = V_i$$

KCL at v_-

$$\frac{V_i - 0}{R_1} + \frac{V_i - V_o}{R_f} = 0$$

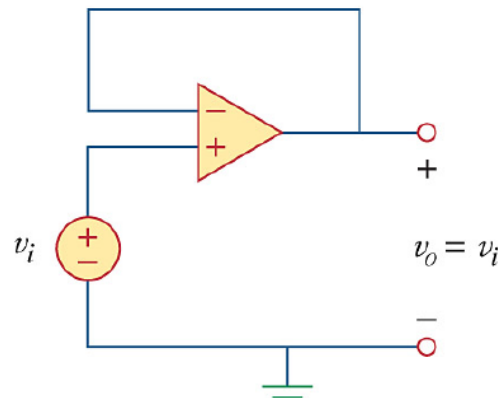
$$\frac{V_i - V_o}{V_i} = -\frac{R_f}{R_1}$$

$$1 - \frac{V_o}{V_i} = -\frac{R_f}{R_1}$$

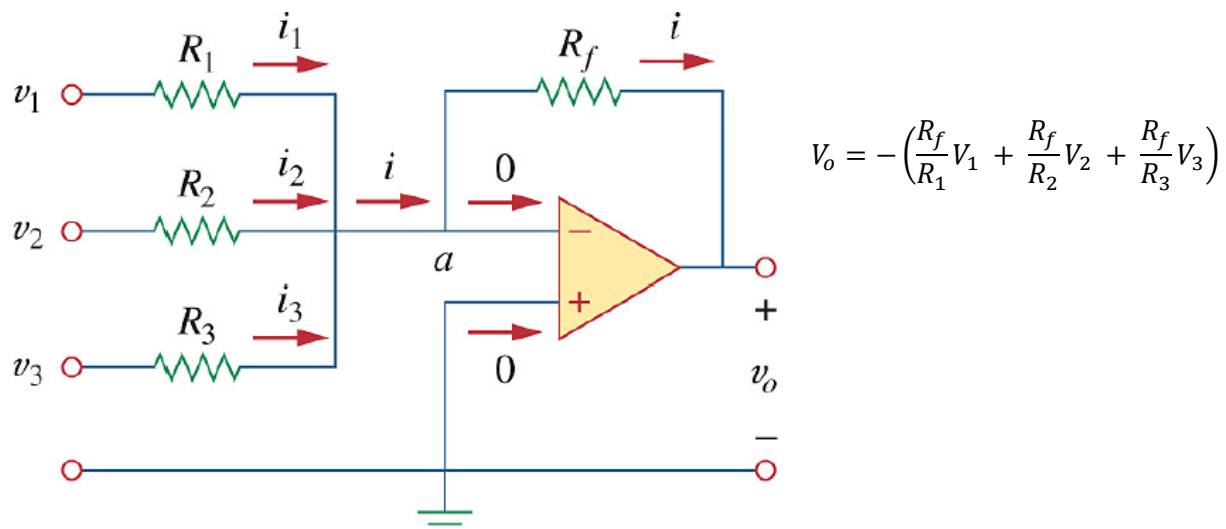
$$\frac{V_o}{V_i} = 1 + \frac{R_f}{R_1}$$

$$V_o = \left(1 + \frac{R_f}{R_1} \right) V_i$$

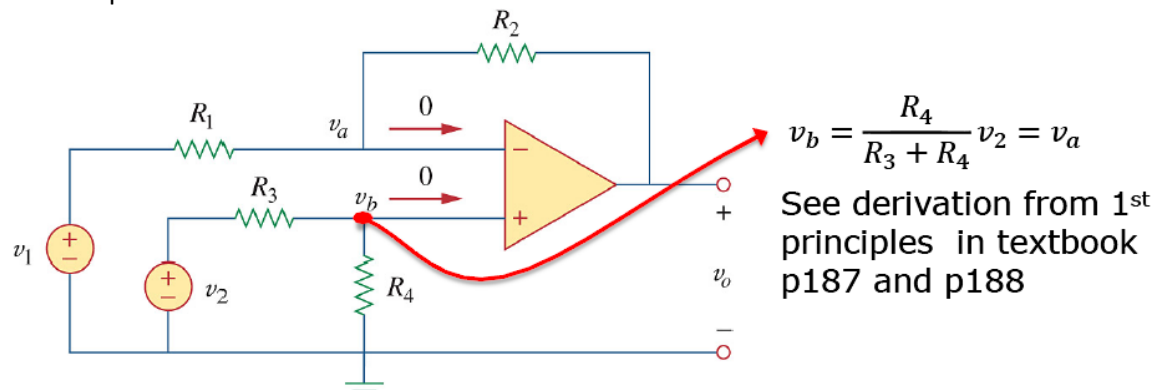
Buffer:



Summing Amplifier:



Difference Amplifier:



$$v_o = \frac{R_2(1 + R_1/R_2)}{R_1(1 + R_3/R_4)} v_2 - \frac{R_2}{R_1} v_1$$

1. If $R_1/R_2 = R_3/R_4$: $V_o = 0$ if $V_1 = V_2$
2. If $R_2 = R_1$ and $R_3 = R_4$: $V_o = V_2 - V_1$

Chapter 6

Capacitors:

$$C = \frac{\epsilon A}{d} \quad [F]$$

$$C = \frac{q}{v}$$

$$i = C \frac{dv}{dt}$$

$$v = \frac{1}{C} \int_{t_0}^t i(t) dt + v(t_0)$$

$$w = \frac{1}{2} C v^2 = \frac{q^2}{2C}$$

$$p = vi = C v \frac{dv}{dt}$$

Series: $\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \dots + \frac{1}{C_N}$

Parallel: $C_{eq} = C_1 + C_2 + \dots + C_N$

Inductors:

$$L = \frac{N^2 \mu A}{l} \quad [H]$$

$$v = L \frac{di}{dt}$$

$$i = \frac{1}{L} \int_{t_0}^t v(t) dt + i(t_0)$$

$$w = \frac{1}{2} L i^2$$

$$p = vi = L \frac{di}{dt} i$$

Series: $L_{eq} = L_1 + L_2 + \dots + L_N$

Parallel: $\frac{1}{L_{eq}} = \frac{1}{L_1} + \frac{1}{L_2} + \dots + \frac{1}{L_N}$

Chapter 9

$$v(t) = V_m \sin(\omega t + \varphi)$$

$$T = \frac{2\pi}{\omega}$$

$$f = \frac{1}{T}$$

$$\omega = 2\pi f$$

$$\begin{aligned} \sin(\omega t \pm 180^\circ) &= -\sin(\omega t) \\ \cos(\omega t \pm 180^\circ) &= -\cos(\omega t) \end{aligned}$$

$$\begin{aligned} \sin(\omega t \pm 90^\circ) &= \pm \cos(\omega t) \\ \cos(\omega t \pm 90^\circ) &= \mp \sin(\omega t) \end{aligned}$$

Rectangular: $z = x + jy = r(\cos\varphi + j\sin\varphi)$

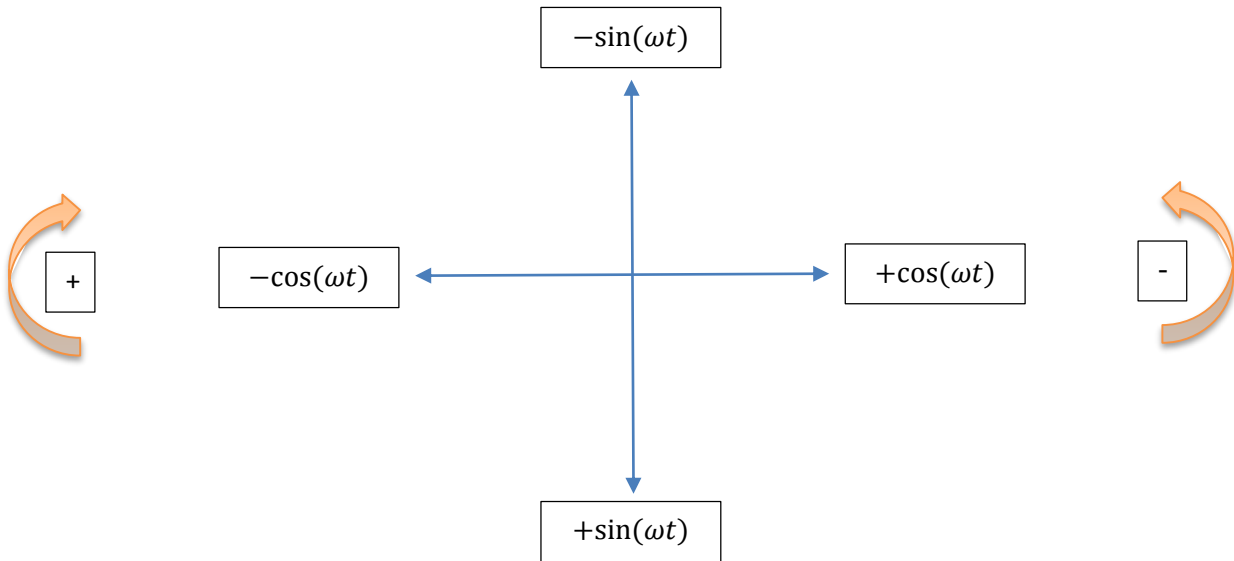
Polar: $z = r \angle \varphi^\circ$

Exponential: $z = r e^{j\varphi}$

$$\frac{1}{j} = -j$$

$$r = \sqrt{x^2 + y^2}$$

$$\varphi = \tan^{-1} \left(\frac{y}{x} \right)$$



Arithmetic:

Addition (+): MUST WORK IN RECTANGULAR FORMAT

$$z_1 + z_2 = (x_1 + x_2) + j(y_1 + y_2)$$

Subtraction (-): MUST WORK IN RECTANGULAR FORMAT

$$z_1 - z_2 = (x_1 - x_2) + j(y_1 - y_2)$$

Multiplication (×): WORK IN POLAR FORMAT

$$z_1 z_2 = r_1 \angle \varphi_1^\circ \cdot r_2 \angle \varphi_2^\circ = r_1 r_2 \angle (\varphi_1^\circ + \varphi_2^\circ)$$

Division (÷): WORK IN POLAR FORMAT

$$\frac{z_1}{z_2} = (r_1 \angle \varphi_1^\circ) / (r_2 \angle \varphi_2^\circ) = \frac{r_1}{r_2} \angle (\varphi_1^\circ - \varphi_2^\circ)$$

Root ($\sqrt{}$): $\sqrt{z} = \sqrt{r} \angle \left(\frac{\varphi}{2}\right)$

Inverse ($\frac{1}{z}$): $\frac{1}{z} = \frac{1}{r} \angle (-\varphi)$

Conjugate (z^*):
$$\begin{aligned} z^* &= (x + jy)^* = (x - jy) \\ &= r \angle (-\varphi) \\ &= e^{-j\varphi} \end{aligned}$$

Euler's identity: $e^{\pm j\varphi} = \cos\varphi \pm j \sin\varphi$

(time domain)	$\leftrightarrow$	(phasor domain)
$v(t) = V_m \cos(\omega t + \varphi) \leftrightarrow$		$\bar{V} = V_m \angle \varphi$
$i(t) = I_m \cos(\omega t + \varphi) \leftrightarrow$		$\bar{I} = I_m \angle \varphi$

Element	Time domain	Frequency domain	Impedance	Admittance
R	$v = Ri$	$\bar{V} = R\bar{I}$	$Z = R$	$Y = \frac{1}{R}$
L	$v = L \frac{di}{dt}$	$V = j\omega LI$	$Z = j\omega L$	$Y = \frac{1}{j\omega L}$
C	$i = C \frac{dv}{dt}$	$V = \frac{I}{j\omega C}$	$Z = \frac{1}{j\omega C}$	$Y = j\omega C$

L: DC Short circuit $\omega = 0, \quad Z = 0$
 High f Open circuit $\omega = \infty, \quad Z = \infty$

C: DC Open circuit $\omega = 0, \quad Z = \infty$
 High f Short circuit $\omega = \infty, \quad Z = 0$